BOOK VI — COSMOLOGY

> *WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md`; subject to chapter-by-chapter audit; nothing here banks.*

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

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VI.0 · Why this sector is treated ruthlessly

Cosmology is where the responsive-vacuum program made its most attractive-sounding claims, and it is the sector where the program's own audit apparatus did the most damage to them. In the committed record this sector contains: two retracted sign claims (one in each direction), a headline exclusion significance retired as impossible in its own channel ($\sim 32\sigma \rightarrow \sim 2.0\sigma$), a "wrong equation of state" verdict on the mechanism proposed to rescue an evolving dark energy, and a standing fence that names the import of an unsourced second timescale for what it is — laundering. The corpus charter therefore requires that this book be built the way the register itself came to be built: the sourced statement first, the inserted structure priced, the retractions on their face, and the ΛCDM comparison stated as what it is — compatibility at a chosen point, not correspondence evidence.

The one-sentence summary, which everything below unpacks: **GRUT's sourced cosmology is ΛCDM.** The framework's only fully-sourced dark-energy statement is $w = -1$ flat; its linear cosmology reproduces ΛCDM at the chosen projector point; every evolving-dark-energy shape requires an inserted, unsourced input; and the sector's genuine products are two directional no-gos (the $\mu = 4/3$ self-exclusion and the kernel-class no-crossing discriminator), a pair of derived structural results about the vacuum kernel (the $s = 5$ spectral law and the exact dS constant tail), and one live external threat (the DESI phantom-divide crossing) that runs *against* the framework. The PREDICTED set is empty here as everywhere.

> **STATUS: EMPTY (nothing has earned entry; Book IX governs entry)** — canonical claim 21, restated for this sector: no cosmological claim in this book carries the label PREDICTED (source: `books/CORPUS_CHARTER.md`, `GRUT_PROGRAM_FREEZE.md`).

VI.1 · Background cosmology

VI.1.1 The background is an input, not a product

GRUT does not derive the background universe. The de Sitter-like late-time background on which every cosmological computation in the record runs is a declared input.

> **STATUS: ASSUMPTION (EMPIRICAL-INPUT — "the dS-like background where used", per the freeze ledger's EMPIRICAL-INPUTS row)** — (source: `GRUT_PROGRAM_FREEZE.md` §3).

Nor does the framework determine the cosmological constant. The register carries this as a marked-open field, deliberately distinguished from both "borrowed input" and "GRUT fill":

> **STATUS: UNRESOLVED (register: tier `to-derive`, sub_status "open field; GRUT does not determine Lambda — marked-open, not claimed")** — the value of Λ is undetermined by the responsive-vacuum framing; the cosmological-constant problem is inherited, posed, and not filled (source: `provenance/claims.json` node `lambda_undetermined`).

What the background *does* yield, given the declaration, is its temperature:

$T_{dS} = H/2\pi$.

> **STATUS: DERIVED (within declarations: forced by Hadamard/KMS on the declared background) —** canonical claim 3; Gibbons–Hawking, borrowed with zero freedom; the register's only background-sector credit (source: `books/CORPUS_CHARTER.md`; `GRUT_PROGRAM_FREEZE.md` §3).

VI.1.2 The sourced background equation of state: w = −1, flat

The framework's admitted gravitational response is pure spin-2 transverse-traceless (the `p_tt_ansatz` projector, with the scalar-to-tensor admixture set at $x = 0$). The spin-0 / trace sector is the one that carries a bulk (volume) response; the TT projector annihilates it. Consequently $\zeta_{bulk} = 0$, the dissipative pressure correction $\Pi = -3\zeta H$ vanishes, and $(w+1) = \Pi/\rho = 0$ — at *background* order, not merely at linear order. The vacuum sits exactly at the de Sitter equilibrium $w = -1$ and stays there.

> **STATUS: DERIVED (within the choices x = 0 / pure-TT: the sourced cosmology statement) —** canonical claim 13, first half; the derivation chain is `p_tt_ansatz` → $\zeta_{bulk} = 0$ → $(w+1) = 0$ at background order (source: `calc/wz_sign.py`, `calc/RESULTS_wz_sign.md` §(iii); `provenance/claims.json` node `rung7_w2_wa_sign`).

The two choices this statement rests on are themselves registered inputs, and their statuses travel with the claim:

> **STATUS: ASSUMPTION (STRUCTURAL-SELECTION — CHOSEN, unanimous five-angle interrogation; not forced) —** canonical claim 8, the TT-only projector: diffeomorphism invariance stops one condition short of forcing tracelessness; the 2026-08-02 interrogation returned CHOSEN (source: `provenance/claims.json` node `p_tt_ansatz`; `BRIEF_p_tt_interrogation.md`).
>
> **STATUS: ASSUMPTION (STRUCTURAL-SELECTION; x_no_pin: the action carries a family, pins nothing) —** canonical claim 9, the $x = 0$ scalar-to-tensor choice: the admissible set is an amplitude-homogeneous cone; excluding the $x = 1$ endpoint (§VI.2.2) does not select the $x = 0$ point (source: `provenance/claims.json` nodes `x_no_pin_theorem`, `mu_linear`).

So the honest reading of "GRUT's background cosmology is ΛCDM" is: *given* an empirically input background, a chosen projector, and a chosen point in an unpinned family, the framework reproduces the cosmological constant it was handed. It also inherits, undischarged, the cosmological-constant problem (`calc/RESULTS_wz.md` §A states this directly: the UV-cutoff single-pole vacuum "reproduces Λ, inherits the cosmological-constant problem").

VI.2 · Linear perturbations and structure formation

VI.2.1 Linear cosmology = ΛCDM at the chosen point

At the chosen point $x = 0$ (pure-TT), the linear-cosmology bookkeeping — modified-Poisson $\mu(a,k)$, lensing $\Sigma(a,k)$, gravitational slip $\eta = \Psi/\Phi$ — collapses to the ΛCDM values exactly: $\boldsymbol{\mu = \Sigma = \eta = 1}$**.** Growth of structure, lensing, and the ISW effect are then standard; GRUT adds nothing and subtracts nothing at linear order.

> **STATUS: DERIVED (conditional leg — GIVEN the CHOSEN $c_0 = 0$ / $x = 0$; register tier `derived-pending`, sub_status `no_go_export`; empirically SELECTED by exclusion, not derived-clean)** — the register is explicit that "μ = 1 likely IS right, but NOT because 'TT annihilates scalars'"; it survives because the alternative endpoint is excluded (§VI.2.2), which is a no-go export, not a positive derivation (source: `provenance/claims.json` node `mu_linear`; `calc/mu_linear.py`, bookkeeping only).

Two pieces of register history are part of this claim's honest presentation. First, the graduation route "derive the scalar-sector vanishing from the action without inserting P^TT" was run in 2026-08-02 and **answered against** (the projector is CHOSEN); the only surviving graduation route relocates the assumption to the rung3 trace correlator at the price of a new +1 — relocation, not discharge. Second, the register carries, at the owner's order, a verbatim dissent against its own tier: the argument that "a conditional on a demonstrably arbitrary choice is assumed-conditional, not derived-pending," recorded precisely because *linear cosmology = ΛCDM is the register's most-quoted headline* and a stale tier at this node would do more downstream laundering work than anywhere else. An armed tier trigger stands: if rung3 resolves against the trace-correlator route, the node moves `derived-pending` → `assumed` without re-litigation (source: `provenance/claims.json` node `mu_linear`, `boundary_condition`).

VI.2.2 The μ = 4/3 self-exclusion — the sector's genuine earned no-go

The modification GRUT's own conformal coefficient would naively suggest is the trace-only endpoint $\mu = 1 + \alpha = 4/3$. The framework *excludes its own naive suggestion*:

> **STATUS: CLOSED (self-exclusion: separate-universe consistency + low-ℓ ISW)** — canonical claim 14, verbatim (source: `books/CORPUS_CHARTER.md`; `provenance/claims.json` node `mu_linear`; `calc/isw_exclusion.py`, `calc/RESULTS_isw_exclusion.md`).

This is the sector's one genuinely earned deflationary result, and its computed anatomy matters because its history is a model case of the program's discipline working on itself:

- **The structural leg (separate-universe consistency).** A super-horizon adiabatic mode is a shifted FRW background whose comoving-gauge growing mode ($\delta \propto a$ in EdS) is fixed by the Friedmann equation; super-horizon $\mu = 4/3$ forces $\delta \propto a^{1.186}$ — internally inconsistent, dataset-independent. EdS-quantified 2026-08-03: $p(4/3) - p_{SU} = +0.186 \neq 0$. Scored **usable-but-conditional** — the owed residue (adiabaticity + the presupposed dilatation bridge, which the L0 screen scored "relocated, not derived") is named, not hidden (source: `calc/RESULTS_isw_exclusion.md`, Part 3 / PART E).
- **The empirical legs.** ISW–galaxy cross-correlation: **computed ~2.0σ** (1.97 central, Σ-corrected; kill-condition band ~0.6–4.8). DESI Σ_0 lensing: ~3.5σ, independent; joint ~4σ-class. The low-ℓ TT auto-power channel is a **prospect, not a leg** (estimate-grade, order-10^2σ-class at $x = 1$, filter/normalization-sensitive; its rigorous calc is owed).
- **The retired number.** The previously banked ~32σ ISW exclusion is **retired**: for a signal-suppressing model the cross-channel exclusion is structurally capped at ~9–12σ, so a 32-class number was never possible in that channel; and the banked mechanism was backwards ($\mu > 1$ *strengthens* growth and suppresses potential decay — the model suppresses the positive ΛCDM-like signal to ~0.57× the template; the exclusion operates because the data detect the positive

signal). The correction and the same-wave firewall's Σ-factor repair (B1) are both on the record (source: `calc/RESULTS_isw_exclusion.md`).

> **STATUS: REVERSED (the banked ~32σ and its mechanism direction; retired 2026-08-03 by `calc/isw_exclusion.py`; the exclusion itself SURVIVES multi-leg at re-graded strength)** — part of model history, stated on its face (source: `calc/RESULTS_isw_exclusion.md`; `provenance/claims.json` node `mu_linear`, `overturning_computation`).

The signature-audit classification of the surviving result is exact: this is **signature-removing**, not predictive. "GRUT *forbids* the $\mu = 4/3$ modification its own coefficient naively suggests; linear cosmology = ΛCDM." It removes a would-be signature and predicts no new one (source: `SIGNATURE_AUDIT.md`, audit table row 4; `NO_GO_LEDGER.md` entry 2).

VI.2.3 The interior window — bounded by a measurement, not deleted by fiat

Because the projector is CHOSEN rather than forced, the continuous {shear, bulk} family *between* the endpoints ($x = 0$, TT-only, $\mu = 1$; $x = 1$, trace-only, $\mu = 4/3$, excluded) is empirically live. The 2026-08 interior wave replaced "closed by fiat" with a computed window: the DESI Σ_0 lensing bound binds at **$x <$ ~0.59** (central-inputs, loose-upper per the named F-MAP fence), i.e. **$\mu - 1 \lesssim 0.20$**, $\Sigma - 1 \lesssim 0.10$; the owed TT-auto calculation would likely re-tighten this substantially.

> **STATUS: UNRESOLVED (register tier `to-derive`, default-BROKEN; "constrained-to-a-computed-window" — the signature audit's fourth category; x has NO FLOOR, so no detection confirms GRUT and no null refutes it: the family allows up to the edge and predicts nothing)** — (source: `provenance/claims.json` node `zeta_interior_family`; `SIGNATURE_AUDIT.md` audit table row 6 and "fourth category" note; `calc/RESULTS_isw_exclusion.md`, binding-inversion row).

The honesty point the audit insists on: the window does **not** soften the signature-null verdict. Structure formation in GRUT is ΛCDM structure formation at the chosen point, with a data-bounded, floorless allowance for departure that carries no predictive content.

VI.3 · Dark energy: the sourced prediction, and the hypothesis that must not be

promoted

VI.3.1 The sourced statement

Repeating VI.1.2 in dark-energy language, because this is the sector's backbone: the sourced prediction of GRUT-as-written is **$w = -1$, flat, at all redshifts**. This is signature-null — it is ΛCDM (source: `SIGNATURE_AUDIT.md`, audit table row 2: "signature-null (sourced = ΛCDM)").

VI.3.2 Evolving w(z): a hypothesis with a price tag

A finite-memory vacuum has a frequency-dependent $\chi(\omega)$; out of equilibrium, its effective w(z) can in principle leave −1. The record explored this as a differentiator and found the structure, priced straight (`calc/wz_dark_energy.py`, `calc/RESULTS_wz.md`):

- w(z) deviates from −1 only if the vacuum has response power at $\omega \sim H(z)$. The confirmed UV-cutoff memory scale gives $H_0\tau\_c \sim 10^{-40}$: **w = −1 flat to ~80 decimals.** No evolution.
- Observable evolution therefore **requires a second, cosmologically slow scale $\tau_2 \sim 1/H_0$ —** exactly the "second internal bath scale" the single-pole spine forbids, coexisting with it only via a ~40-order scale separation (a two-scale vacuum, an explicitly named structural commitment).
- The de Sitter horizon forces the noise temperature and the *existence* of relaxation, but **not the timescale**: both examined horns *insert* $\tau_2 \sim 1/H$ (a free field needs a tuned m ~ H — the η-problem; a self-interacting mode relocates the insertion into a structural bundle). The candidate economical rescue — the rung-9 conformalon doing double duty as the IR mode — is **DEAD, frozen as a disposition**, on three grounds with prefactors carried: its pinned stress is w = +1/3 (radiation-like — the wrong equation of state for a dark-energy deviation), the w-deviation lands ~8× below DESI's amplitude at N = 60, and no Starobinsky–Yokoyama dynamical mass exists for the Δ_4 Paneitz compensator (source: `provenance/claims.json` node `rung7_wz`, `boundary_condition`; `calc/RESULTS_conformalon.md`, cited there).

> **STATUS: HYPOTHESIS (requires the inserted, un-sourced $\tau_2 \sim 1/H_0$, priced +2)** — canonical claim 13, second half, verbatim; the register's ledger for `rung7_wz` is +3 total, of which the τ_2 commitment constitutes +2 and the single-departure-shape closure the remaining +1 (source: `books/CORPUS_CHARTER.md`; `provenance/claims.json` node `rung7_wz`, `ledger_note`).

The laundering fence is the operational form of this status. The forward-model harness carries the invariant **evolving ⇒ needs_unsourced_input**: across the responsive-medium kernel family, the only clean data-consistent vacuum is w = −1 ΛCDM, and *every* DESI-like evolving signal the machine can produce is flagged and refused on inserted-input grounds — independent of the sign of the evolution (the harness's own DESI-sign representative "matches DESI's sign within precision and is still refused"). Importing τ_2 silently, or re-billing an evolving w(z) as a GRUT prediction, is the specific act the fence exists to catch (source: `SIGNATURE_AUDIT.md`, audit-critic verdict; `calc/RESULTS_wz_sign.md`, harness-showcase note).

VI.3.3 The w_a-sign history: two retractions, told on their face

The record's most instructive cosmological episode is not a result but a double retraction.

1. **The first over-claim (away from DESI).** The 2026-06-25 toy computed $w_a > 0$ for the simplest passive relaxor and the record briefly carried "$w_a > 0$ — the WRONG sign for DESI." Retracted 2026-06-29 as un-earned: it was only one of two passivity-consistent branches, and the toy's slope was a ζ = const/Eckart modeling artifact.
2. **The mirror over-claim (toward DESI).** The overseer's own sharpening — "$w_a \le 0$ is second-law-fixed" — was then run through an independent workflow *by its own author* and retracted the same day (the verify-the-verifier principle operating on the verifier).

What survived both retractions is a precise partition: **the second law fixes the SIDE, not the SLOPE.** Passivity ($\zeta \geq 0$) puts the dissipative branch at $w \leq -1$ and the reactive branch at $w \geq -1$, and the entropy production $\sigma = \Pi^2/(\zeta T) \geq 0$ forbids a within-branch crossing of -1 — that is the robust content. But σ is quadratic in Π and blind to $d\Pi/dt$, so the w_a slope rides on sign[$d(\zeta H)/da$], which the dissipation inequality never touches: ζ = const gives $w_a < 0$, $\zeta \sim 1/H^2$ gives $w_a > 0$, both fully passive.

> **STATUS: UNRESOLVED (the w_a sign is genuinely indeterminate; both directional claims — "$w_a > 0$ wrong sign" and "$w_a \leq 0$ second-law-fixed" — RETRACTED 2026-06-29; two non-theorem arguments lean $w_a < 0$, banked as notes, not tiered claims)** — with the additional side-tension recorded: the dissipative branch's $w \leq -1$ floor sits on the *wrong side* of DESI's present-day $w_0 > -1$, so the natural-reading lean lost, on the side axis, even its direction (source: `provenance/claims.json` node `rung7_w2_wa_sign`; `calc/RESULTS_wz_sign.md`, "Overseer rulings" section; `calc/wz_sign.py`).

VI.3.4 The no-crossing no-go: robust, and correctly held below its ceiling

The robustly supported derived-candidate in the w(z) story is the **no-crossing** statement: a single passive channel's deviation ($w+1$) is one-signed and shrinks toward the de Sitter attractor, so w approaches -1 from one side and never crosses the phantom divide. A crossing requires the equilibrium itself off -1 (a real quintessence/phantom degree of freedom) or a sign-changing kernel (≥2 modes / oscillatory poles) — exactly the inserted structure the laundering fence prices.

The register nevertheless **holds this at `to-derive`**, by explicit overseer ruling, for two reasons that this corpus must not smooth over: (1) the no-go is *generic* — it is Vikman 2005 (arXiv:astro-ph/0407107; a single non-ghost degree of freedom cannot dynamically cross $w = -1$), and GRUT's specific content is only "the single-pole spine lands in that class"; (2) it is *conditional on rung3, which is open* — the one-signed-Π argument needs a single real pole, and if the vacuum bath is collisionless free-streaming (a branch cut), the argument fails and the no-crossing needs re-derivation. **A no-go cannot outrank its anchor.**

> **STATUS: UNRESOLVED (held at `to-derive`, overseer-ruled 2026-06-29: generic-flavored — Vikman 2005 cited at any shipped strength — and conditional on the open rung3 anchor; the would-be export `rung7_w3` is contingent and default-BROKEN)** — (source: `provenance/claims.json` nodes `rung7_w2_wa_sign`, `rung7_w3_nocrossing_export`; `calc/RESULTS_wz_sign.md`).

And the anchor itself, for the reader tracking the dependency:

> **STATUS: UNRESOLVED (anchor-class, derived-pending; pole-vs-cut open; the Tier-4 computation found a CUT, not a pole, at flat scope)** — canonical claim 4, verbatim; rung3 is the named bottleneck for the entire w(z) story (source: `books/CORPUS_CHARTER.md`; `provenance/claims.json` node `rung3_single_pole`).

VI.4 · The kernel-class discriminator — what the sector actually exports

The 2026-08-23 discriminator computation (`PHYSICS_LEDGER/rung7_discriminator.py`, `PHYSICS_LEDGER/RUNG7_TWO_POLE_COMPARISON.md`) answered the question "what property of the kernel actually controls the crossing?" with a frozen pre-scan criterion and planted controls, and the answer sharpened the whole export:

- Single-pole Debye: no crossing. Two real poles: no crossing. Three real poles: no crossing. One-channel non-Debye Cole–Cole (a branch-cut kernel): no crossing.
- A damped **oscillatory pole pair**: TRUE CROSSING.

The operative variable is the existence of a second dynamical mode with independent phase (oscillation) — not pole count, and not single-pole structure. Every purely relaxational kernel stays on one side of $w = -1$; single-pole is *decorative* for this observable (the registered derivation never invokes χ's spectral form).

> **STATUS: DERIVED (class-level; explicitly not GRUT-specific)** — canonical claim 15, verbatim: no purely relaxational kernel crosses $w = -1$; only an oscillatory pole pair does. Menu-scope exclusion shared by the entire passive class (source: `books/CORPUS_CHARTER.md`; `PHYSICS_LEDGER/RUNG7_TWO_POLE_COMPARISON.md`, `PHYSICS_LEDGER/rung7_discriminator.py`; consistency-reproduction of Vikman 2005, not a discovery).

The consequence cuts both ways, and the freeze document elevates the adverse direction to a standing reopening condition: **any observed $w = -1$ crossing excludes the entire passive relaxational family at once — GRUT included — at a stroke.** This is the sector's real falsifiable content: a direction, shared with a whole class, not a signature owned by GRUT (source: `GRUT_PROGRAM_FREEZE.md` §5, reopening condition 4; `SIGNATURE_AUDIT.md`).

VI.5 · The DESI anti-signature — the live threat

The 2026-08-02 external hunt (pre-registered A/B/C bar, adversarially refereed, load-bearing numbers overseer-verified against primary literature) produced two cosmology-relevant items, recorded in `SIGNATURE_AUDIT.md` and restated here at their audited strength.

Item 1 — the scoped null. No candidate differentiator above Grade C exists *in the explored structure* (the admitted TT-channel / passive-KMS / ΛCDM-at-linear family, across the four searched domains). This confirms the audit's EMPTY verdict from outside — and it is explicitly *not* the universal claim "GRUT is empirically silent," which no search can establish (source: `SIGNATURE_AUDIT.md`, 2026-08-02 item 1).

Item 2 — the DESI anti-signature, with four attribution fences. GRUT's sourced prediction is $w = -1$ flat; the passivity no-go forbids a single passive relaxor from crossing $w = -1$. DESI's preferred w_0w_a trajectory **crosses the phantom divide at $z \approx 0.35$–0.5** — exactly the forbidden shape. The fences, verbatim from the audit (overseer-verified):

> **(a)** the preference is DESI **BAO + CMB + SNe** — DESI BAO alone shows no significant preference, and with a fixed r_d anchor the ~3σ does not reproduce; **(b)** the honest headline is **3.1σ (DESI DR2+CMB, arXiv:2503.14738)** — the 4.2σ endpoint rides the contested DESY5 compilation (live Efstathiou-vs-DES systematics dispute; one reanalysis drops it to 0.5–1.5σ); **(c)** Bayesian model comparison on the same data gives only weak-to-moderate evidence; **(d)** the direction is Quintom-B — *phantom in the past, quintessence today* (secondary summaries routinely state it backwards).
>
> **STATUS: pending refutation if the signal consolidates** — the audit's verbatim standing disposition, with its comparative clause carried: "GRUT is structurally worse-placed to survive it than ΛCDM+quintessence" (source: `SIGNATURE_AUDIT.md`, 2026-08-02 item 2).

The logical geometry deserves one plain paragraph. The framework's cleanest class-level derived result (VI.4) and its most probable near-term empirical test point at the same observable from opposite sides. If the DESI-preferred crossing consolidates, the whole passive relaxational family — GRUT's admitted structure among it — is excluded without any GRUT-specific measurement ever being needed. If it dissolves, ΛCDM stands and GRUT's sourced cosmology remains indistinguishable from it. There is no branch in which the sector delivers a confirming signature; the audit's phrase for this shape is *anti-signature*.

VI.6 · Derived structural results in the cosmological sector

Three results in this sector carry the word DERIVED at frozen scope, and one canonical reversal frames them. None yields a number an experiment can chase — that, too, is on the record (the admissible set is an amplitude-homogeneous cone; "every route from this framework to a number runs outside it," `GRUT_PROGRAM_FREEZE.md` §3) — but they are the sector's genuine mathematical content.

VI.6.1 The spectral law s = 5

The flat-scope one-loop TT kernel of the gravitational vacuum, at contract scope (ω ≫ H), is

```
Σ_R(ω>0) = −(3/1280π²) ω⁴ [log(μ²/ω²) + iπ] + H²(−(13/480π²)) ω² [log(μ²/ω²) + iπ] + lo
```

giving Im $\chi \sim \omega^4$ as an exact power law on the flat slice — **s = 5 in the registered J-convention** ($J \sim \omega^5$), convergent, with Re $\chi(0) = 3/(2560\pi^2)$ exactly. The physics headline is deflationary in the program's characteristic way: the framework's **own registered s = 3 is rejected** by the frozen tolerance — the two-derivative TT-TT-TT vertex contributes ω^4 in $|V|^2$ on the gapless cut, which the original rung3 density-of-states argument never counted. The registered ω^3 power re-enters only as the curvature-induced $O(H^2)$ component, with an H^2-proportional coefficient the registered family excludes ("H_dependence: NONE declared"). Same side of the decision axis (convergent), structurally different analytic form at leading flat order.

> **STATUS: DERIVED (flat scope; rejects the framework's own registered s = 3)** — canonical claim 6, verbatim; importing s = 3 anywhere downstream is laundering, per the dispatch spec (source: `books/CORPUS_CHARTER.md`; `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`, `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md`; `AGENT_COORDINATION.md`, 2026-09-01 entries).

VI.6.2 The exact de Sitter constant tail

On exact dS_4, the massless minimally coupled field — hence each free TT graviton polarization — has an exact constant zero mode ($\Delta_- = 0$), and the retarded response carries an **exactly constant tail $H^2/4\pi$ filling the interior of the light cone**, verified by a two-sided causality gate sited away from any degenerate point; conformal coupling gaps it at $m\_eff^2 = 2H^2$. This "$\Delta_- = 0$ zero-mode complex" is the structure that survived every deletion test in the resurrection campaign — the record's one surviving fixed point.

> **STATUS: DERIVED (exact dS; gapped only at conformal coupling)** — canonical claim 7, verbatim (source: `books/CORPUS_CHARTER.md`; `RAI_GORILLA_T1.md` §XVI-G; `GRUT_PROGRAM_FREEZE.md` §3).

VI.6.3 What the constant tail did to the finite-memory bet

The two derived results above are not neutral decorations; the second one *reversed the sector's founding intuition*. The finite-memory / single-pole kernel was the program's central asserted object, and the constant tail is its negation in the regime cosmology lives in: what exact dS free-field theory forces is **infinite, scale-free memory** — the opposite shape of the asserted kernel. The one in-house quantitative support for finite memory fell to a bug validated only at its blind point; the prior ("something surely forces finite memory") was removed by enumeration of all seven candidate memory mechanisms.

> **STATUS: ASSUMPTION, with REVERSED history on its face (the in-house "no memory time" computation was reversed; exact dS free-field theory forces infinite scale-free memory)** — canonical claim 18, verbatim; retained as postulate only (source: `books/CORPUS_CHARTER.md`; `RAI_GORILLA_T1.md` §XVI-H; `GRUT_MODEL_FRAMEWORK.md` §7).

VI.6.4 The Γ_T closure — the one cosmological number, computed to refuse

The only parameter-free number this sector ever produced is the closure of the gravitational-wave friction route. The Tier-4 derived kernel's induced cosmological tensor friction is $\Gamma\_T(\omega) = (3/1280\pi)(\omega^3/\bar{M}\_P^2)[1 + (104/9)(H_0/\omega)^2]$ — chromatic, μ-independent — evaluating to **$\Gamma\_T/H_0 = 6.19\times10^{-63}$ at 100 Hz: 62.7 orders below the shared-slot bound** few×H_0. The pre-registered question ("does the local memory scale connect to the cosmological friction parameter-free?") is answered: no parameter-free connection exists on the licensed record, and the SPEC's own gate returns REFUSE on the observable route.

> **STATUS: CLOSED (computed NO EFFECT; SPEC outcome REFUSE on the observable route; commits 2116251, 41e1af5) —** canonical claim 16, verbatim (source: `books/CORPUS_CHARTER.md`; `calc/RESULTS_gw_tensor_friction.md`, `calc/gw_tensor_friction.py`, `GRUT_PREDICTION_GATE_GAMMA_T.md`).

VI.7 · The ΛCDM comparison, stated whole

Assembling the sector: at background order the sourced statement is $w = -1$ flat (VI.1.2); at linear order $\mu = \Sigma = \eta = 1$ at the chosen point (VI.2.1); the one naive departure the framework's own coefficient suggests is self-excluded (VI.2.2); the admissible interior departure is data-bounded with no floor (VI.2.3); every evolving-dark-energy shape is priced as an inserted input (VI.3.2); and the class-level no-crossing direction is shared with all passive media (VI.4). The comparison with ΛCDM is therefore not "GRUT reproduces ΛCDM" in the correspondence-limit sense that phrase usually carries. It is: *GRUT-as-written, at its chosen point, is observationally ΛCDM*, and the choices that put it there are registered assumptions. Per the freeze's mandatory honesty note, recovery of standard results by standard machinery on declared inputs is **compatibility, not correspondence evidence**, and nothing in this sector counts toward PREDICTED.

What the sector genuinely contributes, at its earned strengths:

product	kind	status label carried
$w = -1$ flat	sourced statement	DERIVED (within the choices $x = 0$ / pure-TT)
$\mu = 4/3$ endpoint exclusion	self-exclusion (no-go export)	CLOSED
no purely relaxational kernel crosses $w = -1$	class-level theorem-grade result	DERIVED (class-level; not GRUT-specific)
$s = 5$ spectral law; dS constant tail $H^2/4\pi$	structural results about the vacuum kernel	DERIVED (at their frozen scopes)
evolving $w(z)$	hypothesis, priced	HYPOTHESIS (+2, τ_2 un-sourced)
DESI crossing	external anti-signature	pending refutation if the signal consolidates
Γ_T at $\omega \sim 100$ Hz	closure	CLOSED (computed NO EFFECT)

VI.8 · Absence map — where the record is silent

The corpus charter treats an absence map as valid content. The following standard-cosmology topics have **no GRUT account in the frozen record**. Listing them here is a statement about the record, not a promise; under the stopping rule, none may be filled by invention.

> **STATUS: UNMAPPED —** for every item below (the label is canonical claim 22's for its listed sectors; extended here descriptively to sector-adjacent silences, each verified silent against the register and the ledgers).

- **Dark matter and baryogenesis** — canonical claim 22, verbatim: "Flavor, strong-CP, neutrino masses, dark matter, baryogenesis — **UNMAPPED**." The register's 74 nodes contain no GRUT account of the dark-matter sector or the baryon asymmetry.
- **Inflation and the early universe.** No GRUT inflationary mechanism, spectrum, or initial-condition account exists in the record. (E-fold counts appear only inside the conformalon closure arithmetic, VI.3.2; the Past Hypothesis input is a time-orientation datum, not an early-universe model.)
- **BBN, recombination, reionization, CMB physics beyond the low-ℓ ISW/TT channels used in the μ = 4/3 exclusion.** Nothing in the record.
- **Nonlinear structure formation** (halo formation, N-body regime, baryonic feedback). The record's structure-formation content is entirely linear-order bookkeeping.
- **The observational tensions** (H_0, S8). No GRUT position exists; no node addresses them.
- **The ω ≲ H regime** — not merely unmapped but **UNASKABLE at current declarations, on four obstructions** (not false — unposable); this is where the noise-sector fork and the white-floor regime live, outside the contract truncation (source: `GRUT_PROGRAM_FREEZE.md` §3 UNRESOLVED row; `AGENT_COORDINATION.md` 2026-09-01).
- **Black-hole ringdown/QNM** — the signature audit's one flagged soft spot: expected invisible by inheritance of the rung4 Planck suppression, but this is an inheritance argument, **not a dedicated computation**; the calc does not exist (source: `SIGNATURE_AUDIT.md`, "the one soft spot").

Sources drawn from

- `books/CORPUS_CHARTER.md` (canonical status table; formatting)
- `GRUT_MODEL_FRAMEWORK.md` (authoritative presentation; §5 cosmology, §7 failures)
- `GRUT_PROGRAM_FREEZE.md` (ledger; stopping rule; reopening conditions)
- `provenance/claims.json` — nodes `rung7_wz`, `rung7_w1_wz_map`, `rung7_w2_wa_sign`, `rung7_w3_nocrossing_export`, `mu_linear`, `rung3_single_pole`, `zeta_interior_family`, `lambda_undetermined`, `vc_w_equals_minus_one`, `p_tt_ansatz` (via node texts)
- `SIGNATURE_AUDIT.md` (items 1–2 of the 2026-08-02 hunt; audit table; fourth category; QNM soft spot)
- `NO_GO_LEDGER.md` (entries 2 and 3; strength legend)
- `calc/wz_dark_energy.py` + `calc/RESULTS_wz.md`
- `calc/wz_sign.py` + `calc/RESULTS_wz_sign.md`
- `calc/mu_linear.py` (bookkeeping, via register node)
- `calc/isw_exclusion.py` + `calc/RESULTS_isw_exclusion.md`

- `calc/RESULTS_gw_tensor_friction.md` (+ `calc/gw_tensor_friction.py`, `GRUT_PREDICTION_GATE_GAMMA_T.md`)
- `PHYSICS_LEDGER/RUNG7_TWO_POLE_COMPARISON.md` (+ `PHYSICS_LEDGER/rung7_discriminator.py`)
- `PHYSICS_LEDGER/WALL_KR_CONTRACT_BENCHMARK_VERDICT.md`, `PHYSICS_LEDGER/WALL_KR_CONTRACT_RETARDED_VERDICT.md` (via `AGENT_COORDINATION.md` 2026-09-01 consolidated entries)
- `RAI_GORILLA_T1.md` (§XVI-G, §XVI-H: the constant tail; the memory reversal)
- External literature, as already cited by the record: Vikman 2005 (arXiv:astro-ph/0407107); Gubitosi–Piazza–Vernizzi (arXiv:1210.0201); DESI DR2+CMB (arXiv:2503.14738); DESI MG (arXiv:2411.12026); Li–Barrow (arXiv:0902.3163); Salcedo–Colas–Dufner–Pajer (arXiv:2507.03103); Calzetta–Hu (book); Kubo 1966.

Gaps in this book

1. **The register's authority-vocabulary annotation applies throughout.** Words like "overseer," "specialist," "referee," and "external" in the sources quoted here denote in-house passes — separate AI sessions run by the program's one human author; no outside human has been contacted (sealed audit `PREREG_AUTHORITY_TERMS_2026-08-12.txt`, quoted in `rung7_wz` and `rung3_single_pole`). This book inherits that vocabulary when quoting; the reader must not read it as independent review.
2. **The owed calculations are owed, not done**: the low-ℓ TT-auto exclusion calc (the gate for interior viability above $x \sim 0.06$); the separate-universe residue (adiabaticity + the dilatation bridge); the de Sitter trace-sector effective stress tensor that would decide the w_a slope; the rung3 transport self-energy that anchors the no-crossing; the dedicated QNM/ringdown calc.
3. **This book does not re-derive the $s = 5$ kernel or the dS tail**; it restates the frozen verdicts and their scopes from the Wall-A/K_R ledger artifacts. The full contract chain (declarations D1–D5, the scheme fork, the T4 adjudication queue) belongs to Books II/IV.
4. **The DESI empirical situation is quoted as of the record's 2026-08-02 verification**; nothing newer is imported, per the charter's no-new-external-claims rule.
5. **No account of the {shear, bulk} interior's dynamics** ($\mu(c_0)$, slip, stability) exists in the record beyond the window arithmetic; the register holds it at `to-derive`, default-BROKEN, and this book adds nothing to it.
6. **Background cosmology beyond the dS-like patch** (matter era dynamics, radiation era, transitions) has no GRUT-specific account; the record's $w(z)$ toys run on an input ΛCDM background $H(a)$, and this book flags rather than fills that dependence.